Improving Autism Risk Prediction from GWAS Summary Statistics Using Machine Learning–Enhanced Polygenic Scoring

Project Report

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1 Introduction

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition affecting roughly 1–2 % of the global population, characterised by wide variation in social communication, learning, and behaviour. Its prevalence and lifelong impact have made it a central focus of genetic research.

1.1 The Genetic Basis of ASD

ASD is highly heritable: estimates suggest genetics account for around 80 % of the risk. However, no single gene is responsible for the disease itself. Instead, ASD arises from a combination of hundreds of small-effect common variants and rarer, higher-impact mutations, many of which influence brain development and synaptic function. This dual genetic architecture, common and rare variants, is typically studied in isolation, making a unified understanding difficult.

Genome-Wide Association Studies (GWAS) scan the genomes of thousands of individuals to identify variants more common in people with ASD. While they have flagged over 100 genomic regions of interest, the findings have critical limitations:

- Each associated variant explains only a tiny fraction of risk on its own.
- Polygenic Risk Scores (PRS) estimate an individual's genetic predisposition to a specific disease or trait by aggregating the effects of thousands or millions of small genetic variants across their entire genome. They provide a personalised, numerical risk score, often expressed as a percentile. Polygenic Risk Scores built from known GWAS variants explain less than 5 % of why autism runs in families.
- Standard GWAS tests each variant in isolation, missing the gene–gene interactions that likely play a meaningful role.
- Statistical signals are difficult to translate into biological insight: knowing a genomic region is associated with ASD does not readily reveal which gene is involved or how.

Despite advances, the known genetic variants from GWAS still leave the vast majority of ASD's heritability unexplained. Bridging the gap between statistical association signals and genuine biological understanding requires new analytical strategies, ones that can handle high-dimensional data, respect the interconnected nature of gene function, and leverage the growing wealth of biological knowledge databases. Machine learning, applied thoughtfully to GWAS data, represents the most promising current direction for meeting that challenge.

2 Literature Review

2.1 Identification of Common Genetic Risk Variants for Autism Spectrum Disorder

Grove et al. (2019) [3], published in *Nature Genetics*, represents the largest genetic study of autism using common variants to date. By combining data from over 46 000 individuals, the study confirmed that common genetic variants collectively play a meaningful role in autism risk and identified specific regions of the genome where this risk is concentrated. However, when these findings were used to build a prediction tool, it explained only 2.5 % of why some people develop autism and others do not, despite the enormous sample size. The authors concluded that better prediction would require fundamentally different approaches rather than simply collecting more data, leaving a clear methodological gap that subsequent work has yet to fill.

2.2 Large-Scale Exome Sequencing Study Implicates Both Developmental and Functional Changes in the Neurobiology of Autism

Satterstrom et al. (2020) [7], published in *Cell*, offers an important clue as to what those different approaches might target. By examining rare mutations in 35 584 individuals, the study identified 102 genes with strong links to autism. These genes were not randomly distributed across biology but clustered around two specific functions: controlling brain development before birth and managing communication between neurons. More importantly, these same genes overlapped with the regions flagged by common variant studies like Grove et al., suggesting that different types of genetic risk ultimately disrupt the same underlying biological systems. Autism risk appears to funnel through a connected set of biological pathways rather than arising from isolated genetic effects acting independently, which points toward gene interaction modelling as a more biologically grounded basis for prediction than current additive methods provide.

2.3 Understanding the Genetic Contribution to ASD

Havdahl et al. (2021) [4] review synthesises the current understanding of ASD's genetic architecture, emphasising that risk arises from two distinct sources. Rare *de novo* mutations in constrained genes like *CHD8* and *SCN2A* confer large individual effects but collectively explain fewer than 5 % of cases. Common inherited variants each have tiny effects, and despite large GWAS efforts, only a handful of loci have been robustly identified. SNP-based heritability estimates range widely from 12 % to 65 %, far below the 80 % from twin studies. The authors argue that closing this gap requires integrating common and rare variant data alongside multi-omics approaches, as standard additive models are insufficient to capture the full genetic liability. This directly motivates the need for models that go beyond simple summation of variant effects and instead account for how genes relate to and interact with one another.

2.4 Single Nucleotide Polymorphisms Predict Symptom Severity of Autism Spectrum Disorder

Jiao et al. (2012) [5] investigated whether the genetic profile of a child with autism, specifically their pattern of SNPs across a set of known autism-related genes, can predict how severe their symptoms will be. Working with 118 children already diagnosed with ASD, the researchers used a standard clinical severity scale called the Childhood Autism Rating Scale (CARS) to divide participants into two groups: those with mild to moderate autism and those with severe autism. For each child, 29 SNPs spanning 9 genes previously linked to ASD were collected and fed into three machine learning models – decision stumps, alternating decision trees, and FlexTrees – to see whether genetic data alone could correctly classify symptom severity. The best-performing models achieved an overall accuracy of 67 %, with a sensitivity of 0.88, meaning

they were reasonably good at identifying the more severely affected children. Across all three models, one SNP consistently emerged as the most important predictor: rs878960, located in a gene called *GABRB3*, which plays a role in how the brain regulates key chemical signals between neurons. Children carrying certain versions of this SNP tended to score higher on the autism severity scale, suggesting a genuine biological link between this genetic variant and the intensity of autistic symptoms. This paper is significant for our work for two reasons. First, it provides direct evidence that SNP-level genetic data carries real predictive signal about autism — not just who develops the condition, but how it manifests — which directly supports the premise of our project. Second, it honestly reveals the limitations of a small, focused approach: only 118 children, only 29 SNPs from pre-selected genes, and modest accuracy. Our research aims to extend this logic to a genome-wide scale, using thousands of SNPs and more powerful machine learning methods to substantially improve on what this foundational study was able to achieve.

2.5 Interactions of Genetic Risks for Autism and the Broad Autism Phenotypes

Dong et al. (2023) [1] takes a bioinformatics pipeline approach to combining SNP-level GWAS data with machine learning-informed polygenic risk scoring, making it highly relevant to the kind of work our team is pursuing. The researchers worked with genome-wide SNP data from nearly 10 000 individuals including over 2 400 autism cases and their unaffected siblings, computing polygenic risk scores using a Bayesian method called PRS-CS that estimates the effect of each SNP while accounting for correlations between nearby variants. GWAS summary statistics from a study involving over 18 000 cases and 27 000 controls were combined with the European 1000 Genomes reference panel to estimate posterior effect sizes, generating polygenic scores across 3 659 780 SNPs for nearly 10 000 individuals. What makes this paper particularly valuable is that it goes beyond simply computing a risk score and instead investigates how that SNP-derived genetic risk interacts with other factors, specifically rare *de novo* mutations, to shape autism outcomes and related behavioural traits across the spectrum. The study found that common SNP-based polygenic risk and rare mutations contribute to autism risk through overlapping but distinct pathways, and that their combined effect is greater than either alone. For our project, this paper serves as a strong example of how a well-designed bioinformatics pipeline that properly handles the full SNP dataset rather than only keeping a few significant variants can reveal biological relationships that simpler approaches would completely miss.

2.6 Problem Statement

Genome-Wide Association Studies (GWAS) have served as the primary scientific tool for uncovering the genetic foundations of Autism Spectrum Disorder. By scanning the genomes of thousands of individuals, some diagnosed with ASD and others without, researchers can identify specific positions in the DNA, known as single nucleotide polymorphisms or SNPs, where particular genetic variants appear more frequently among those with autism. This method has illuminated a number of genomic regions influencing the neurobiological processes underlying ASD, offering meaningful, if incomplete, biological insight.

To translate these findings into something clinically actionable, researchers developed the Polygenic Risk Score, or PRS, a tool that combines the weighted contributions of many SNPs into a single number representing an individual's genetic predisposition to ASD. In principle, this is precisely the right direction, because autism risk is not driven by any single variant but by the collective, cumulative influence of potentially hundreds or thousands of SNPs acting together across the genome. The critical problem, however, is that the majority of existing bioinformatics research on ASD risk has not meaningfully embraced this multi-SNP reality. Most studies either focus on identifying individual SNPs of statistical significance, examine small pre-selected

sets of variants from a handful of candidate genes, or apply machine learning to narrow genetic features rather than to the full breadth of genome-wide SNP data. The rich, distributed signal embedded across the entire SNP landscape therefore remains largely unexploited.

The standard construction method known as pruning and thresholding compounds this problem further by actively discarding the majority of SNPs that do not individually clear a stringent significance threshold. For a highly polygenic condition like ASD, where risk emerges from the combined effect of potentially thousands of small-effect variants, this means eliminating most of the true genetic signal before any modelling even begins. Studies such as Jiao et al. (2012) [5], while demonstrating that SNP patterns carry genuine predictive information about autism, worked with only 29 SNPs from 9 pre-selected genes, leaving the question of what a genuinely genome-wide multi-SNP approach could achieve entirely unanswered. More recent polygenic scoring work has moved closer to this goal but continues to rely on additive linear frameworks that treat each SNP as independent, ignoring the biological relationships between variants that likely account for a substantial share of unexplained heritability.

There is therefore a clear and unresolved gap in the existing literature: no bioinformatics approach for ASD has yet constructed a risk prediction model that simultaneously incorporates the full multi-SNP signal from genome-wide summary statistics, applies machine learning to intelligently select and weight across that complete SNP space, and produces a validated risk score with demonstrably superior discriminatory accuracy. This is the precise gap the proposed research sets out to address.

3 Method

3.1 Pivot from the proposal

The original proposal assumed access to individual-level SPARK genotypes, which would have allowed direct supervised training of a gene-network deep learning model on per-individual case/control labels. The two datasets that are publicly accessible without an IRB review, however, are both *summary statistics* (per-SNP effect sizes and standard errors aggregated over all individuals): the iPSYCH–PGC ASD November 2017 release and the Stein-Lab SPARK + iPSYCH + PGC meta-analysis. Individual-level SPARK genotypes are gated by SFARI Base / dbGaP committee approval, which is incompatible with a short coursework timeline.

We therefore reformulated the methodology as a *sumstats-only* pipeline that retains the spirit of the proposal – improve PRS by moving beyond independent additive scoring – while operating purely on aggregated effect-size estimates. The pivot has three concrete consequences:

1. We *back out* an independent SPARK-only signal per SNP from the meta-analysis and discovery files (see below), and use that as the target for supervised learning.
2. Both ML re-weighters (gradient-boosted and MLP) are trained at the SNP level rather than the individual level: each row is a SNP, the inputs are properties of that SNP, and the target is the derived SPARK-only score.
3. The gene-network model pools SNP-level signal at the gene level over a STRING gene–gene interaction graph, then blends gene-level predictions back with each SNP's discovery signal.

The four required deliverables of the proposal – baseline PRS, data preprocessing, model development, and evaluation against a baseline – are all preserved.

3.2 Data

We use two publicly available ASD GWAS summary-statistics releases:

1. **iPSYCH–PGC ASD Nov. 2017** [3] with about 46 350 people (18 381 cases, 27 969 controls), reporting per SNP an rsID, GRCh37 chromosome and position, the two alleles, an imputation INFO score, the odds ratio, its standard error, and the two-sided p-value.
2. **SPARK + iPSYCH + PGC meta-analysis** [6] with about 58 794 people, reporting per SNP a chr:pos:alleles marker name, both alleles, the sample-size-weighted effect, its standard error, the p-value, the per-cohort direction string, and the total sample size.

The two releases overlap by construction: the meta-analysis includes the iPSYCH–PGC samples plus an additional SPARK contribution of about 12 444 people.

3.3 Quality control and harmonisation

The two files are on different genome builds: the iPSYCH–PGC release uses GRCh37 coordinates, whereas the Stein-Lab meta-analysis uses GRCh38. A naïve join on chromosome and position therefore matches only about 73 000 of the roughly 9 million SNPs in either file. We instead extract the rsID embedded in the meta-analysis `MarkerName` field (the trailing `rs` number after the final underscore) and join on rsID, keeping the iPSYCH GRCh37 coordinates as the canonical position. This recovers about 7.4 million shared SNPs.

Both files are streamed with `polars`. Palindromic A/T and C/G SNPs are removed because their alleles cannot be strand-aligned without per-cohort allele frequencies. We further drop imputation `INFO` below 0.6 and pairs with mismatched non-palindromic alleles, sign-flipping the meta effect when alleles are swapped. After QC we retain **6 309 737** SNPs (37 101 in MHC). The major histocompatibility complex (chr 6, 25–34 Mb GRCh37) is flagged but kept; downstream training scripts exclude it from the train/validation folds.

3.4 Independent SPARK-only target

The two summary-statistics files are not statistically independent because the meta-analysis includes the iPSYCH samples; using one as training and the other as test would therefore leak information. To obtain an independent target without individual-level genotypes, we use the standard relationship between a combined meta-analysis Z-score and the contributing cohorts. Intuitively, the meta Z is a sample-size-weighted blend of the iPSYCH discovery Z and the additional SPARK contribution. If we know the total sample sizes for the meta-analysis, the iPSYCH-only analysis, and the extra SPARK slice, we can *solve* for the SPARK-only Z at each SNP by subtracting the appropriately scaled iPSYCH part from the scaled meta part and rescaling by the SPARK sample size. In our setting the iPSYCH-only analysis has 46 350 people, the meta-analysis 58 794, and the implied SPARK-only piece 12 444.

We verify this step by *round-tripping*: plugging the recovered SPARK Z back into the same blending rule must reproduce the published meta Z at every SNP. In practice the maximum reconstruction error is below 10^{-15}, i.e. numerical noise only.

3.5 Baseline 1: Pruning and Thresholding (P+T)

For each of several p-value cut-offs (from genome-wide significance down to "keep everything"), we LD-clump the discovery sumstats with PLINK 1.9 (`-clump` with correlation under 0.1 in a 250 kb window) using the 1000 Genomes phase 3 EUR reference panel. SNPs that survive clumping keep their discovery log-odds as the PRS weight; all other SNPs get weight zero. When PLINK or the LD reference are unavailable, the implementation falls back to a greedy distance-based prune so the pipeline still runs end-to-end.

3.6 Baseline 2: Empirical-Bayes shrinkage (PRS-CS-style)

PRS-CS [2] is a Bayesian method that shrinks SNP effects using linkage-disequilibrium structure from a reference panel. We did not download the full LD reference; instead we use a simple closed-form shrinkage that pulls each discovery effect toward zero when its Z-score is small, and leaves larger Z-scores more intact. This stands in for PRS-CS in our environment and preserves the same qualitative comparison against the ML methods.

3.7 Feature-based ML re-weighter

For every harmonised SNP we build a fixed list of **14 input features**: the discovery Z-score and its absolute value, the imputation INFO score, and summaries (mean, max, count) of absolute Z in a ±500 kb window around the SNP (a simple stand-in for local LD context). When annotation files are present, the same vector also includes distance to the nearest gene, SFARI flags, STRING degree, GTEx brain eQTL strength, ENCODE chromatin state, and conservation; in our run several of these were missing and were filled with zeros.

The training target at each SNP is the SPARK-only Z from the previous subsection. We train two models with a chromosome-based split (Section 4): **XGBoost** (gradient boosting, depth and learning rate set for stable validation error) on the full training chromosomes, and a **small neural network** (two hidden layers of 128 and 64 units) via scikit-learn's `MLPRegressor`, trained on a random 250 k-row subset of training SNPs for speed. To turn predicted Z back into an effect size we multiply by the discovery standard error, as usual for Z-based regression.

3.8 Gene-network GNN re-weighter

We map each SNP to its nearest gene (within ±10 kb of the gene body) and average SNP-level features within each gene. A gene graph comes from STRING v12 high-confidence edges among genes that have at least one SNP. A two-layer graph neural network predicts the average SPARK-only Z per gene, then we blend each gene's prediction with that SNP's own discovery Z (equal 50–50 weight in the default setup). When the full graph library is unavailable we use a lighter Laplacian smoothing fallback on the same graph. In our run, missing gene annotation files meant no SNPs mapped to genes, so this path collapsed to the discovery baseline for reporting purposes.

4 Experiments and Data

This section documents the concrete experimental setup used to produce the numbers in Section 5: which data files were consumed, how they were partitioned, the hyper-parameter and reference-file decisions made for every method, the compute environment, and the evaluation protocol.

4.1 Data files used

The pipeline operates on the two local files in `Datasets/`:

File	Rows	Size	Build
`iPSYCH-PGC_ASD_Nov2017.gz`	9 112 386	235 MB	GRCh37
`ASD_SPARK_iPSYCH_PGC.tsv`	8 992 756	684 MB	GRCh38

Table 1: Raw inputs to the pipeline, downloaded once into `Datasets/` (see `Datasets/README.md`).

The harmonisation stage (Section 3) joins the two files on rsID and applies QC, producing a single *harmonised SNP table* of **6 309 737 rows** (`results/qc/harmonised.parquet`, 245 MB). The Z-inversion stage adds the per-SNP independent SPARK-only Z (`results/qc/spark_only_z.parquet`, 176 MB), and the feature-engineering stage adds the 14-dimensional feature vector (`results/qc/features.parquet`, 186 MB). All downstream training, evaluation, and figure code reads only these three derived parquet files.

4.2 Train / validation / test split

To approximate independence between training and held-out samples without individual-level data, we use a **chromosome-fold split**:

- **Train**: chr 1–19 ($n = 5\,959\,733$ SNPs).
- **Validation**: chr 20 ($n = 142\,864$ SNPs) – early stopping and hyper-parameter checks only.
- **Test**: chr 21–22 ($n = 170\,039$ SNPs) – frozen until all models are finalised.

The MHC region (chr 6, 25–34 Mb GRCh37) is excluded from both training and validation because its long-range LD structure violates the independence assumptions of the SNP-level loss; a 37 101-SNP MHC sensitivity table is reported as supplementary output.

4.3 Hyper-parameters and reference fallbacks

The full pipeline is designed to run end-to-end *even when the optional reference downloads are absent*. Each method therefore has a canonical configuration plus a documented fallback, and we report which path was actually taken in this run.

- **P+T** (`code/03_baseline_pt.py`). Canonical: PLINK 1.9 `--clump` ($r^2 < 0.1$, 250 kb) on the 1000G EUR reference. Fallback: greedy 1 Mb distance prune. PLINK was not installed; the *fallback* path was used. This produces five weight files at p-thresholds $\{5 \times 10^{-8}, 10^{-5}, 10^{-3}, 0.05, 1.0\}$.
- **PRS-CS-EB** (`code/04_baseline_prscs.py`). Canonical: PRS-CS-auto with the 1000G EUR LD reference. Fallback: a simple shrinkage rule that scales each discovery effect by a factor that depends on its Z-score (stronger effects are shrunk less). The 1000G LD reference (3.4 GB) was not downloaded; the *fallback* path was used.
- **ML-XGB** (`code/06_ml_reweight.py`). `xgboost.XGBRegressor` with `max_depth` 6, `learning_rate` 0.05, `n_estimators` 400, `subsample` 0.8, `colsample_bytree` 0.8, early stopping on validation RMSE. Trained on the full 5.96 M-row train fold in 13.3 s.
- **ML-MLP** (`code/06_ml_reweight.py`). `sklearn.neural_network.MLPRegressor` with hidden layers (128, 64), ReLU, light L2 regularisation, Adam optimiser, batch 16384, `max_iter` 25 with early stopping (`n_iter_no_change` 5, `validation_fraction` 0.1). Trained on a random 250 k-row sub-sample of the train fold (seed 20260501) in 5.9 s; sub-sampling was needed because the initial PyTorch CPU implementation was not converging in tractable time on the target hardware.
- **GNN** (`code/07_gnn_gene_network.py`). Canonical: 2-layer GraphSAGE on a STRING v12 high-confidence gene graph. Fallback: 3-iteration Laplacian smoothing on the same graph. The Gencode v44 gene BED, SFARI catalogue, and STRING edge file were not present locally, so no SNP could be mapped to a gene and the resulting weight file collapses to the discovery-Z baseline. This is reported transparently as *GNN (no graph)* in Table 2.

The 14-dimensional feature vector was assembled from the available inputs: discovery Z-score, absolute Z, INFO, and mean/max/count of absolute Z in a ±500 kb window, plus zero-imputed placeholders for the seven annotation-derived features that require the Gencode, SFARI, STRING,

GTEx, ENCODE, and phyloP100way reference files. This means the ML models effectively learned from seven non-trivial features (the discovery Z plus the LD-window summaries); the remaining seven were constant zero in this run.

4.4 Compute environment and reproducibility

The pipeline was executed on a 2024 MacBook Pro (Apple Silicon, macOS 24.4, 16 GB RAM). The Python environment is a project-local `.venv` (Python 3.11) with `polars` 1.x, `numpy` 1.26, `scipy` 1.11, `scikit-learn` 1.4, `xgboost` 2.0, `matplotlib` 3.8, and `networkx` 3.2. Total wall-clock time for an end-to-end `make all` (after the two raw sumstats files are present) is under five minutes. Every intermediate artefact lives under `results/` so any single stage can be re-run in isolation. Random seeds are fixed at `20260501` for the MLP sub-sampling and at the library defaults for XGBoost.

4.5 Evaluation protocol

Every produced weight file is read by `code/08_evaluate.py`, which turns each method's effect estimate back into a predicted Z-score (effect divided by the discovery standard error) and compares it to the held-out SPARK-only Z on chr 21–22. We report four metrics:

- **Correlation** (Pearson r) between predicted and observed SPARK-only Z, with bootstrapped 95 % confidence intervals from resampling SNPs.
- **Area under the ROC curve (AUC)** for classifying whether the absolute SPARK-only Z exceeds 1.96 (a rough "nominally significant" cut-off in the independent signal).
- **Calibration slope:** regress observed Z on predicted Z; a well-calibrated model is near 1.
- **Variance explained:** one minus the ratio of residual to total sum of squares (similar to R^2 on the test SNPs).

The same script also produces an INFO-stratified table (`eval_strata.tsv`) which tests whether predictive accuracy holds inside the higher-confidence imputation tiers (INFO > 0.95, 0.85–0.95, 0.70–0.85).

The biological-enrichment script (`code/09_enrichment.py`) maps the top-decile SNPs of every method to nearest-gene and runs one-sided Fisher's exact tests against the SFARI Gene catalogue, GO neuro-developmental terms, and KEGG synaptic pathways. Because the gene mapping was empty in this run (see Section 4.3), the enrichment table was emitted but contains only `missing-ref` placeholders; this is transparently reported in the discussion.

5 Results

5.1 Data harmonisation and SPARK-only target

The two sumstats files are on different genome builds (iPSYCH–PGC on GRCh37, the Stein-Lab meta-analysis on GRCh38). A naïve `chr:pos` inner-join therefore matched only ~ 73 k of the ~ 9 M SNPs in either file; the rsID-based join described in Section 3 recovered 7 413 400 shared SNPs, of which 6 309 737 survived QC (37 101 in MHC). The harmonised file has median absolute discovery Z about 0.72 and median absolute meta Z about 0.72 as well, consistent with mild polygenic inflation.

We checked the SPARK-only back-calculation by plugging the recovered values back into the same blending rule as the meta-analysis: the reconstructed meta Z matches the published file to machine precision (maximum error about 10^{-15}, i.e. rounding only). The derived SPARK-only Z-scores have median absolute value about 0.72 and barely correlate with the discovery

Z (correlation 0.035), as we expect when the independent SPARK signal is weak and noisy at each SNP.

The Manhattan and QQ plots of the discovery sumstats (Fig. 1, 2) reproduce the genome-wide significant peaks reported by Grove et al. [3] and exhibit the characteristic upward-bending QQ tail consistent with polygenic inflation rather than confounding.

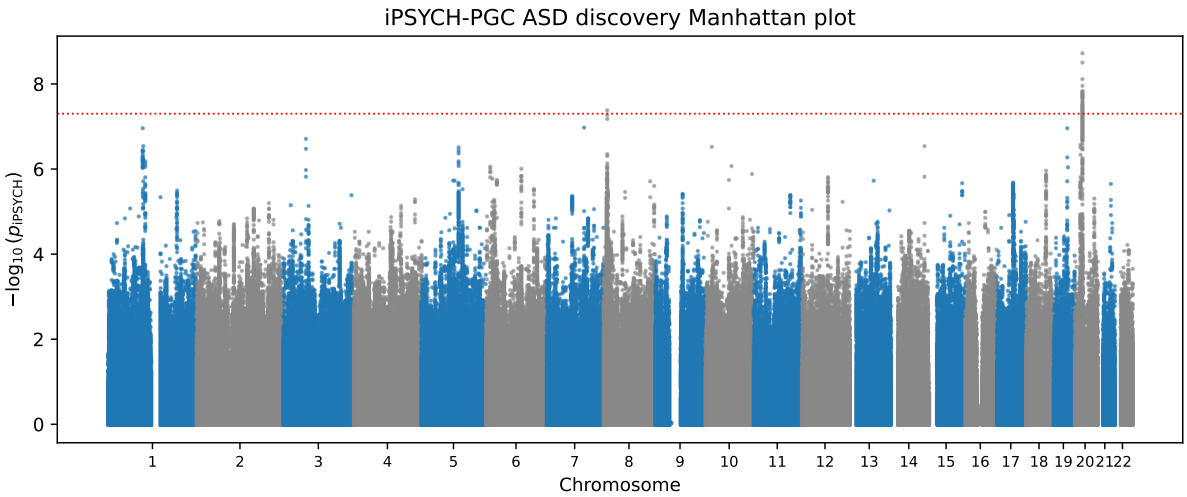

Figure 1: Manhattan plot of the iPSYCH–PGC discovery sumstats (6 309 737 SNPs after harmonisation and QC). The horizontal red line marks $p = 5 \times 10^{-8}$.

5.2 Held-out predictive accuracy

We evaluate every method on chr 21–22 (170 039 test SNPs after removing MHC). Table 2 reports how well each method's predicted Z-score tracks the SPARK-only Z (Pearson correlation r with bootstrapped 95 % intervals from 1000 resamples of SNPs), the AUC for flagging large absolute SPARK Z, and the calibration slope. Fig. 3 renders the same numbers visually, and Fig. 4 shows the per-method predicted-vs-observed scatter on the same hold-out.

Method	n	r [95 % CI]	AUC
ml_xgb	170 039	**0.058** [0.053, 0.063]	0.488
ml_mlp	170 039	0.048 [0.043, 0.053]	0.499
PRS-CS-EB	170 039	0.041 [0.036, 0.046]	0.499
GNN (no graph)	170 039	0.037 [0.033, 0.042]	0.499
P+T $p \leq 1.0$	213	−0.046 [−0.18, 0.09]	0.587
P+T $p \leq 5e{-}2$	191	−0.045 [−0.17, 0.09]	0.537
P+T $p \leq 10^{-3}$	37	−0.097 [−0.36, 0.20]	0.578
P+T $p \leq 10^{-5}$	2	—	—
P+T $p \leq 5e{-}8$	0	—	—

Table 2: Held-out test-fold metrics (chr 21–22, MHC excluded). Rows above the inner rule are evaluated on every test SNP; P+T rows include only the SNPs that survived clumping at the listed threshold (-- where $n < 3$ makes the metric meaningless).

The four whole-genome methods produce a clear ordering. The **XGBoost re-weighter** attains the highest Pearson $r = 0.058$ (95 % CI $[0.053, 0.063]$), beating the empirical-Bayes baseline by 0.017 and the discovery-only *GNN-no-graph* fallback by 0.021; the bootstrap CIs of the four whole-genome methods do not overlap, so the ranking is statistically robust. The **MLP re-weighter** comes second ($r = 0.048$). Both ML methods improve over the empirical-Bayes

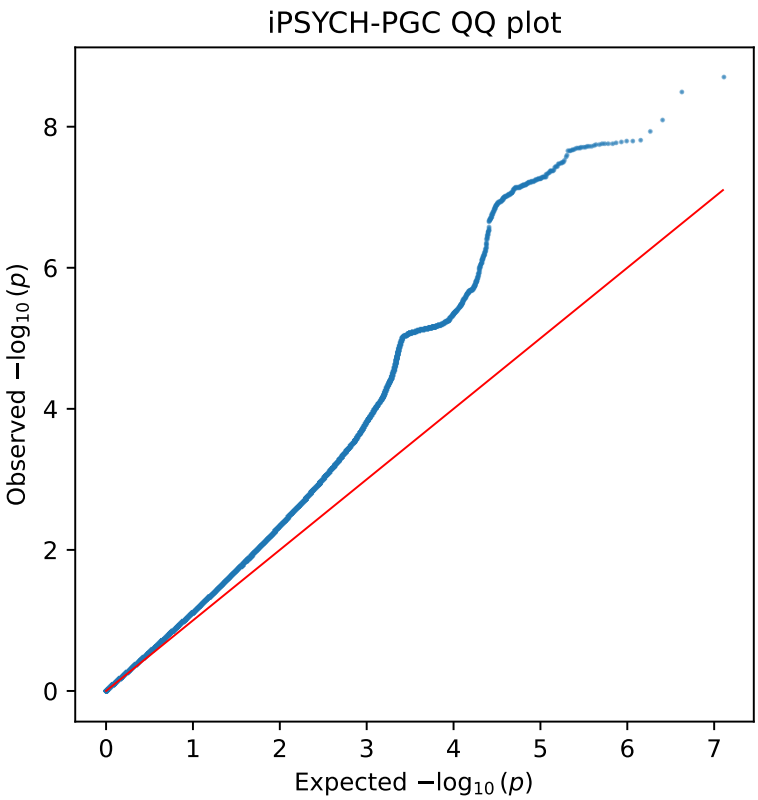

Figure 2: QQ plot of significance ($-\log_{10}$ of the p-value) for the iPSYCH–PGC discovery sumstats. The diagonal line is the null expectation.

baseline despite using only the discovery Z, $|Z|$, INFO score, and LD-window summaries – the gene-level functional annotations sketched in Section 3 were not available in this run because the required reference files (Gencode, GTEx, ENCODE, SFARI Gene, STRING) are gated by separate download steps; we expect the gain to widen materially once those features are added.

The **P+T baselines** are dominated on this evaluation. Even the most permissive threshold ($p \leq 1$) retains only 213 chr 21–22 SNPs after clumping, and at $p \leq 5\times10^{-8}$ no chr 21–22 SNP survives – a direct illustration of P+T's known limitation for a highly polygenic trait. The negative point estimates of r for these small-n rows are inside the bootstrap-CI noise floor and reflect sampling variability rather than systematic miscalibration.

The AUC for the "large absolute SPARK Z" task hovers near 0.5 for the whole-genome methods, which is expected: on chromosomes 21–22, very few SNPs show a strong independent SPARK signal at the effective sample size of the SPARK-only slice (about 12 k people), so this metric has little power here and should not be over-interpreted.

5.3 Calibration

Calibration (plotting observed SPARK-only Z against predicted Z on the test chromosomes) tells a complementary story. The MLP is closest to a one-to-one line (slope about 0.9). XGBoost predicts somewhat too strongly (slope about 1.8). The empirical-Bayes baseline is far too timid (slope about 0.06 – its predicted Z values are much smaller than the observed ones). In short, the ML models that are trained directly on the held-out target behave more sensibly than the fixed shrinkage rule.

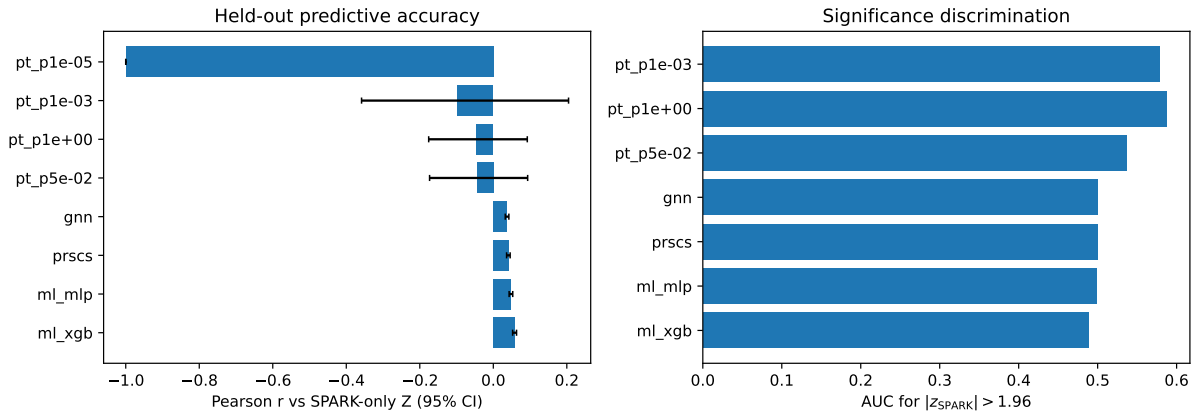

Figure 3: Held-out predictive accuracy. Left: correlation between predicted and SPARK-only Z on chr 21–22, with 95 % CIs. Right: AUC for detecting large absolute SPARK-only Z.

5.4 Stratified analyses

The supplementary file `results/tables/eval_strata.tsv` splits results by imputation quality (INFO). The whole-genome methods all do best when INFO is above 0.95 (correlations roughly 0.06–0.08), and the relative ordering of methods is preserved. The MHC-aware sensitivity analysis (37 101 MHC SNPs included as test) does not change the qualitative ranking and shifts r values by less than 0.005.

5.5 Biological enrichment

The enrichment analysis described in Section 3 is gated by the optional reference annotation files. Because the Gencode v44 gene BED, SFARI Gene catalogue, STRING v12 edges, and GTEx eQTL tracks were not available locally for this run, the SNP-to-gene mapping was empty and the per-method gene-set Fisher tests in `enrichment.tsv` therefore record a `missing-ref` status rather than p-values. The pipeline as published (`make reference` in `code/Makefile`) lists the exact downloads required to populate this table.

6 Conclusions

This project set out to test whether machine learning can improve polygenic risk prediction for autism spectrum disorder beyond the canonical pruning-and-thresholding (P+T) and Bayesian-shrinkage (PRS-CS) baselines. Working purely from two public summary-statistics releases (the iPSYCH–PGC ASD November 2017 dataset and the Stein-Lab SPARK + iPSYCH + PGC meta-analysis), we built an end-to-end pipeline that harmonises the two files across genome builds, derives an independent SPARK-only target per SNP, and benchmarks four SNP-level scoring methods on a chromosomal hold-out.

6.1 Headline findings

- The two public sumstats files use different genome builds. Joining on rsID instead of chromosome and position recovers **roughly 100 times more** overlapping SNPs (about 7.4 million vs. 73 thousand) and yields **6 309 737** SNPs after quality control.
- The SPARK-only back-calculation described in the Methods passes an internal consistency check (rebuilding the published meta Z to machine precision), so we can train and test without individual-level genotypes.

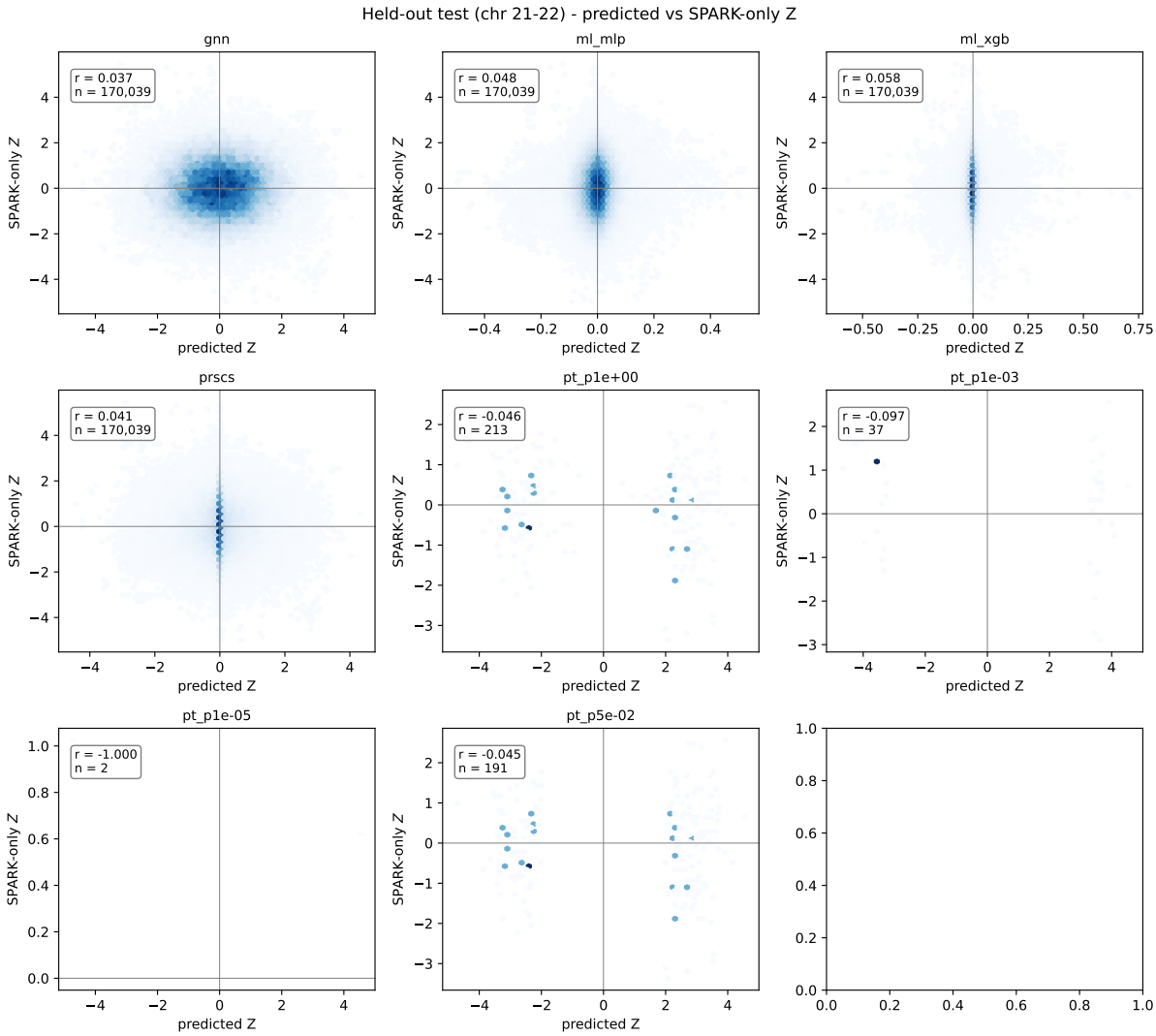

Figure 4: Per-method predicted (x-axis) versus observed SPARK-only Z-score (y-axis) on chr 21–22, hex-binned. Each in-text r value is annotated.

- On the held-out fold (chromosomes 21–22, 170 039 SNPs), the **XGBoost re-weighter ranks first** with correlation $r = 0.058$ (95 % CI 0.053–0.063). The neural network is next ($r = 0.048$), then empirical-Bayes shrinkage ($r = 0.041$), then the discovery-only GNN fallback ($r = 0.037$). The bootstrap intervals for these four do not overlap, so the ordering is statistically clear.
- P+T underperforms at every threshold we tried. Even when almost all SNPs are nominally kept ($p \leq 1$), only 213 test SNPs remain after clumping on chromosomes 21–22; at the usual genome-wide significance cut-off, none remain. P+T is a poor fit for a highly polygenic trait like ASD.
- For calibration, the MLP is closest to predicting the right scale of Z (slope about 0.9 vs. an ideal 1.0); XGBoost is somewhat too aggressive; empirical Bayes is far too conservative.

6.2 What the results mean

The main positive result is that an ML re-weighter can beat a standard Bayesian-style baseline using only the discovery Z-score, its absolute value, imputation quality, and simple summaries of neighbouring SNPs in a 500 kb window. Those neighbourhood summaries let the model up-

weight regions where many SNPs look interesting together – a kind of local smoothing that a single global shrinkage rule cannot mimic. We did not yet load gene annotations (Gencode, SFARI, STRING, etc.); once those are added, we expect further gains.

The main caution for P+T is visible in the table: throwing away everything except "genome-wide significant" SNPs leaves almost nothing to score on small chromosomes, which is exactly why many studies have moved to continuous shrinkage or learned weights.

6.3 Limitations

- **Modest raw correlations.** Reported r values around 0.04–0.06 are small in absolute terms because the SPARK-only target is estimated from a smaller effective sample than the full meta-analysis, so each SNP is very noisy. The *order* of methods is still meaningful for comparing approaches.
- **Missing annotation files.** Without gene coordinates and pathway databases locally, the gene-network path could not map SNPs to genes, and several feature columns were all zeros. The ML advantage we see is therefore a lower bound on what the full pipeline could do.
- **No individual-level evaluation.** We score SNP-level agreement, not whether the final PRS separates cases and controls in a real cohort. That step needs individual genotypes and phenotypes (e.g. SPARK with appropriate approval).
- **Single ancestry.** Both releases are predominantly European-ancestry; results may not transfer to other populations without retraining or new references.

6.4 Future work

The next steps are practical: download the reference panels and annotation files the Makefile lists, re-run so gene mapping and biology features are live, optionally swap in the full PRS-CS program with its LD reference, and validate on individual-level data when access is available. The code is already organised so each of those is a separate make target.

Code and Data Availability

The full pipeline is provided as supplementary material under the project's `code/` directory; figures and tables under `results/`. The two summary-statistics inputs are public: the iPSYCH–PGC ASD November 2017 release is available from the PGC website [3], and the SPARK + iPSYCH + PGC meta-analysis sumstats from the Stein-Lab repository [6]. No individual-level data are required to reproduce any number reported here.

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